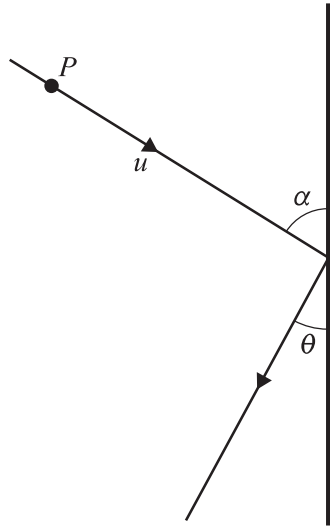




A particle P of mass m is moving with speed u on a fixed smooth horizontal surface. It collides at an angle α with a fixed smooth vertical barrier. After the collision, P moves at an angle θ with the barrier, where $\tan \theta = \frac{1}{2}$ (see diagram). The coefficient of restitution between P and the barrier is e . The particle P loses 20% of its kinetic energy as a result of the collision.

Find the value of e .

[5]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.